

R Code Reference

Estimation I

This handout collects the core R functions used throughout the course, organized by topic. For worked examples, see the companion site.

Probability and Distributions

R Code	Math	Description
<code>dnorm(x, mu, sd)</code>	$\phi(x; \mu, \sigma)$	Normal PDF
<code>pnorm(x, mu, sd)</code>	$\Phi(x; \mu, \sigma)$	Normal CDF
<code>qnorm(p, mu, sd)</code>	$\Phi^{-1}(p)$	Normal quantile function
<code>rnorm(n, mu, sd)</code>	$X \sim N(\mu, \sigma^2)$	Draw n normal random variables
<code>dt(x, df), pt(), qt()</code>		Student t density, CDF, quantile
<code>dchisq(x, df), pchisq(), qchisq()</code>		χ^2 density, CDF, quantile
<code>pf(x, df1, df2), qf()</code>		F distribution CDF, quantile
<code>dbinom(x, n, p), rbinom()</code>		Binomial PMF, random draws
<code>plogis(x), qlogis()</code>		Logistic CDF, quantile
<code>integrate(f, lower, upper)</code>	$\int_a^b f(x) dx$	Numerical integration

Descriptive Statistics

R Code	Math	Description
<code>mean(x)</code>	$\bar{x} = \frac{1}{n} \sum_i x_i$	Sample mean
<code>var(x)</code>	$\frac{1}{n-1} \sum_i (x_i - \bar{x})^2$	Sample variance (Bessel-corrected)
<code>sd(x)</code>	$\sqrt{\widehat{\text{Var}}(x)}$	Sample standard deviation
<code>cov(x, y)</code>	$\frac{1}{n-1} \sum_i (x_i - \bar{x})(y_i - \bar{y})$	Sample covariance
<code>cor(x, y)</code>	$\widehat{\text{Cor}}(x, y)$	Sample correlation
<code>quantile(x, p)</code>		Sample quantiles
<code>length(x)</code>	n	Number of elements

Matrix Operations

R Code	Math	Description
<code>A %*% B</code>	$\mathbf{AB}$	Matrix multiplication
<code>t(X)</code>	$\mathbf{X}'$	Transpose
<code>crossprod(X)</code>	$\mathbf{X}'\mathbf{X}$	Cross product (faster than <code>t(X) %*% X</code>)
<code>solve(A)</code>	$\mathbf{A}^{-1}$	Matrix inverse
<code>solve(A, b)</code>	$\mathbf{A}^{-1}\mathbf{b}$	Solve $\mathbf{Ax} = \mathbf{b}$ (preferred over <code>solve(A) %*% b</code>)
<code>diag(A)</code>	$\text{diag}(\mathbf{A})$	Extract diagonal
<code>diag(n)</code>	$\mathbf{I}_n$	$n \times n$ identity matrix
<code>det(A)</code>	$ \mathbf{A} $	Determinant
<code>dim(A)</code>	$n \times m$	Matrix dimensions
<code>qr(A)\$rank</code>	$\text{rank}(\mathbf{A})$	Matrix rank
<code>eigen(A)</code>		Eigenvalues and eigenvectors
<code>cbind(x, y), rbind()</code>		Bind columns or rows

OLS Regression

R Code	Math	Description
<code>lm(y ~ x1 + x2, data)</code>		Fit OLS regression
<code>coef(mod)</code>	$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$	Coefficient estimates
<code>fitted(mod)</code>	$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta}$	Fitted values
<code>resid(mod)</code>	$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$	Residuals
<code>vcov(mod)</code>	$\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$	Variance-covariance of $\hat{\beta}$
<code>sigma(mod)</code>	$\hat{\sigma} = \sqrt{\mathbf{e}'\mathbf{e}/(n-k)}$	Estimated error std. deviation
<code>confint(mod)</code>		Confidence intervals for $\hat{\beta}$
<code>predict(mod, newdata, interval)</code>		Predictions with optional intervals
<code>summary(mod)</code>		Full regression summary
<code>model.matrix(mod)</code>	$\mathbf{X}$	Extract the design matrix
<code>update(mod, . ~ . + x3)</code>		Add/remove terms from a model

Diagnostics and Sensitivity

R Code	Math	Description
<code>hatvalues(mod)</code>	$h_{ii} = [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']_{ii}$	Leverage (diagonal of hat matrix)
<code>rstudent(mod)</code>		Studentized residuals
<code>anova(mod1, mod2)</code>		F -test comparing nested models
<i>From car:</i>		
<code>linearHypothesis(mod, "x1 = 0")</code>		Wald test of linear restrictions
<i>From sensemakr:</i>		
<code>sensemakr(mod, treatment)</code>		Omitted variable bias sensitivity

Robust and GLS Inference

R Code	Math	Description
<i>From sandwich:</i>		
<code>vcovHC(mod, type = "HC1")</code>		Heteroskedasticity-robust $\widehat{\text{Var}}(\hat{\beta})$ Types: HC0, HC1, HC2, HC3
<i>From lmtest:</i>		
<code>coeftest(mod, vcov = vcovHC)</code>		Test coefficients with robust SEs
<i>From estimatr:</i>		
<code>lm_robust(y ~ x, data, se_type)</code>		OLS with robust SEs in one step

Binary Response Models

R Code	Math	Description
<code>glm(y ~ x, family = binomial(link = "probit"))</code>		Probit regression
<code>glm(y ~ x, family = binomial(link = "logit"))</code>		Logit regression
<code>predict(mod, type = "response")</code>	$\hat{P}(Y = 1 X)$	Predicted probabilities
<code>logLik(mod)</code>	$\ell(\hat{\beta})$	Log-likelihood at MLE
<code>AIC(mod), BIC(mod)</code>		Model selection criteria

Asymptotics and Resampling

R Code	Math	Description
<code>replicate(B, expr)</code>		Repeat expr <i>B</i> times (simulation)
<code>sample(x, n, replace = TRUE)</code>		Bootstrap resampling
<code>set.seed(123)</code>		Set RNG seed for reproducibility
<i>From car:</i>		
<code>deltaMethod(mod, "x1/x2")</code>		Delta method for nonlinear functions
<i>From boot:</i>		
<code>boot(data, statistic, R)</code>		Bootstrap inference
<code>boot.ci(boot.out, type = "perc")</code>		Bootstrap confidence intervals

Instrumental Variables

R Code	Math	Description
<i>From estimatr:</i>		
<code>iv_robust(y ~ x1 z1, data)</code>		2SLS with robust SEs
<i>Manual 2SLS via lm:</i>		
<code>stage1 <- lm(x ~ z)</code>		First stage
<code>lm(y ~ fitted(stage1))</code>		Second stage (SEs require correction)

GMM

R Code	Math	Description
<i>From gmm:</i>		
<code>gmm(g, x, t0)</code>		GMM with moment function g
		<code>g(theta, x)</code> returns $n \times q$ matrix of moment conditions
<code>summary(gmm_mod)</code>		Estimates, SEs, and J -test

Panel Data

R Code	Math	Description
<i>From plm:</i>		
<code>pdata.frame(df, index = c("id","t"))</code>		Declare panel structure
<code>plm(y ~ x, model = "within")</code>		Fixed effects (within) estimator
<code>plm(y ~ x, model = "random")</code>		Random effects estimator
<code>plm(y ~ x, model = "between")</code>		Between estimator
<code>plm(y ~ x, model = "pooling")</code>		Pooled OLS
<code>phtest(fe_mod, re_mod)</code>		Hausman test (FE vs. RE)
<i>From fixest:</i>		
<code>feols(y ~ x id + t, data)</code>		Fast FE with multi-way clustering
<i>Dynamic panels from plm:</i>		
<code>pgmm(y ~ lag(y) + x lag(y, 2:99))</code>		Arellano–Bond / system GMM
<code>sargan(mod), mtest(mod)</code>		Overidentification and autocorrelation tests

Modern Difference-in-Differences

R Code	Math	Description
<i>From did:</i>		
<code>att_gt(y ~ x, data, ...)</code>		Callaway–Sant’Anna group-time ATTs
<code>aggte(att_gt_obj, type = "dynamic")</code>		Aggregate to event-study or overall
<code>ggdid(aggte_obj)</code>		Plot event-study estimates
<i>From fect:</i>		
<code>fect(y ~ x, data, method = "ife")</code>		Interactive FE / matrix completion

Visualization (ggplot2)

R Code	Description
<code>ggplot(data, aes(x, y)) + geom_point()</code>	Scatterplot
<code>+ geom_smooth(method = "lm")</code>	Add regression line with CI band
<code>+ geom_histogram()</code>	Histogram
<code>+ geom_line()</code>	Line plot
<code>+ stat_function(fun = dnorm)</code>	Overlay a mathematical function
<code>+ facet_wrap(~ group)</code>	Small multiples by group
<code>+ labs(x = ..., y = ..., title = ...)</code>	Labels
<code>+ theme_minimal()</code>	Clean theme

Data Manipulation Essentials

R Code	Description
<code>data.frame(x = ..., y = ...)</code>	Create a data frame
<code>sapply(X, FUN), lapply()</code>	Apply a function to each element
<code>tapply(y, group, mean)</code>	Apply a function by group
<code>aggregate(y ~ group, data, mean)</code>	Group-wise summaries
<code>factor(x), cut(x, breaks)</code>	Create categorical variables
<code>do.call(rbind, list_of_dfs)</code>	Stack a list of data frames

Key packages: sandwich, lmtest, estimatr, car, boot, gmm, plm, fixest, did, MASS, ggplot2.